

The psychobiological predictability of experimental asset bubbles

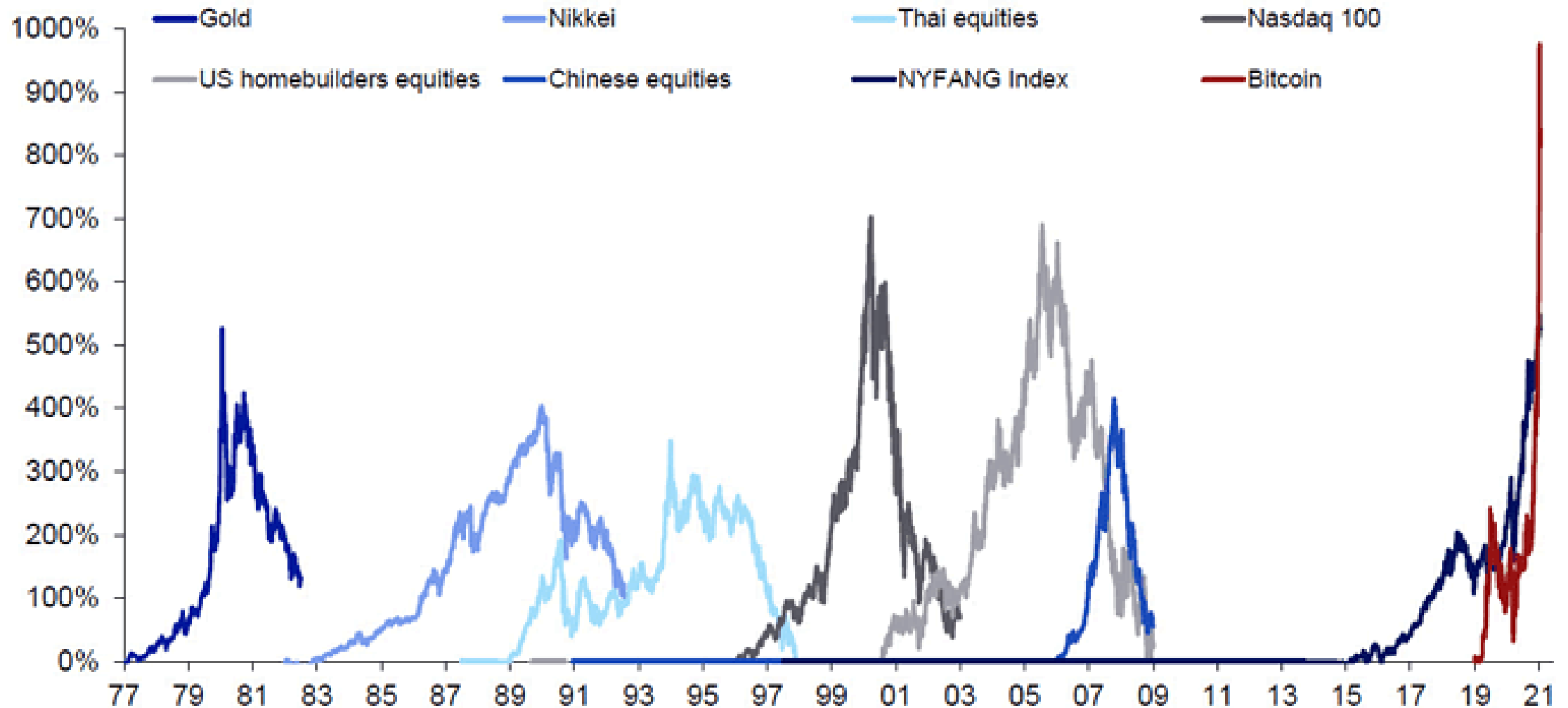
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Introduction

- **Efficient market hypothesis (EMH):** In an efficient market, asset prices fully reflect all available information.
- A **bubble** is an economic cycle that is characterized by the rapid escalation of market value, particularly in the price of assets. This is followed by a quick decrease in value, or a contraction, that is sometimes referred to as a "crash".
- **Effects of bubbles:** misallocation of resources, financial instability, and significant wealth losses when they burst.
- **Rational vs. irrational bubbles:** Rational bubbles occur when investors expect future price increases despite knowing the asset is overvalued; irrational bubbles involve speculative mania without regard to fundamentals.
- Some scholars manifested skepticism with regards to bubbles, arguing that they incorporate rational expectations

45 years of asset bubbles



Difficulty and importance of forecasting



An experimental/neurobehavioral view

- **Difficult to study:** fundamental value, multiple factors affecting pricing
- **Biases:**
 - **Herding:** Investors follow others (crypto – Haykir and Yagli, 2022)
 - **Overconfidence:** Overestimating future returns leads to excessive risk-taking (Camerer, 1989; Michailova and Schimdt, 2016)
 - **Market Sentiment:** Collective emotions drive prices beyond fundamentals, further inflating bubbles (Berger and Turtle, 2015; Osoolian et al., 2021)
- **Neuroeconomics** studies have captured bubble formation and found connection between brain and biometric dynamics with stock returns and behavior (de Martino et al., 2013, Smith et al., 2014, Toma and Miyakoshi, 2021)

Experiments are an excellent tool to understand bubble formation and provide novel forecast predictions

Research questions

- Can biometric measures such as electrodermal activity and heart rate help predict stock returns in a bubble setting?
- Can the same measures be used to forecast bubble crashes?

Objective

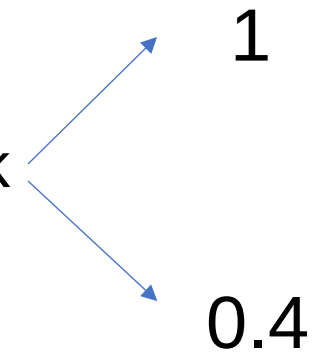
- Run series of experiments and test if biometric data features can forecast stock returns and bubble crashes

Market design (replication Smith et al., 2014)

- Subjects trade for 30 rounds in an open-call market
- Start with 4 stocks and 100 CASH
- At the end, any stock they hold is re-deemed at 14 units regardless of market value

Option 1: Keep cash
for a risk-free return of
5% each round

Option 2: Invest in stock



$$rP_t = E[P_{t+1}] - P_t + E[D]$$

$$P_t = \frac{E[D]}{r} = \frac{0.7}{0.05} = 14$$

Anything above 14 is a bubble

Experimental design

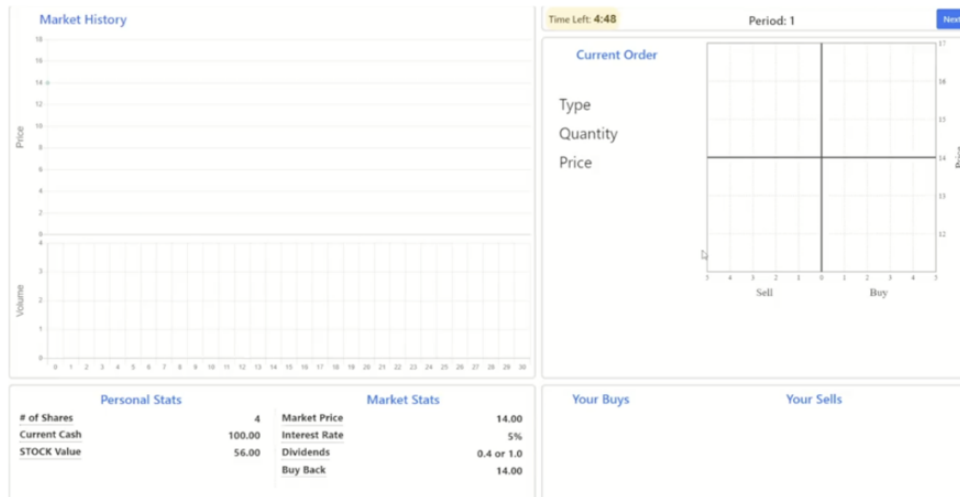
- Caltech SSEL experimental lab
- Empatica E4: EDA, HR, BVP and TEMP
- Markets with avg ~17-23 subjects, total 19 markets
- Avg. 5 students in lab and avg. 15 on **Prolific**



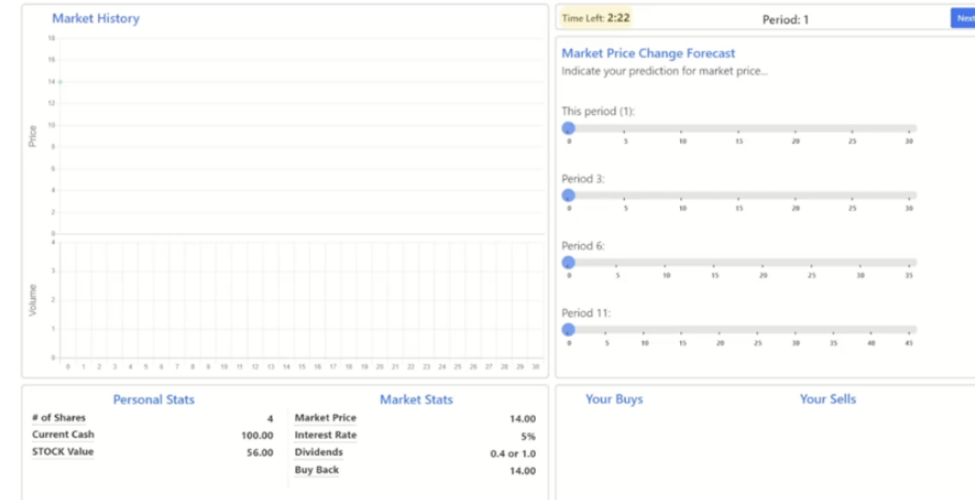
Total duration: ~1.5hrs

Experimental design – sequence

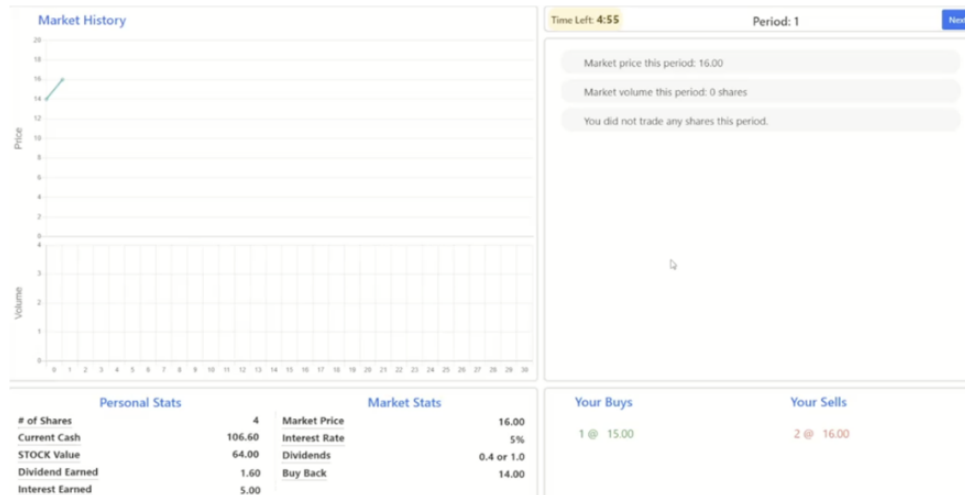
Screen 1: order submission



Screen 2: forecast prices



Screen 3: round results

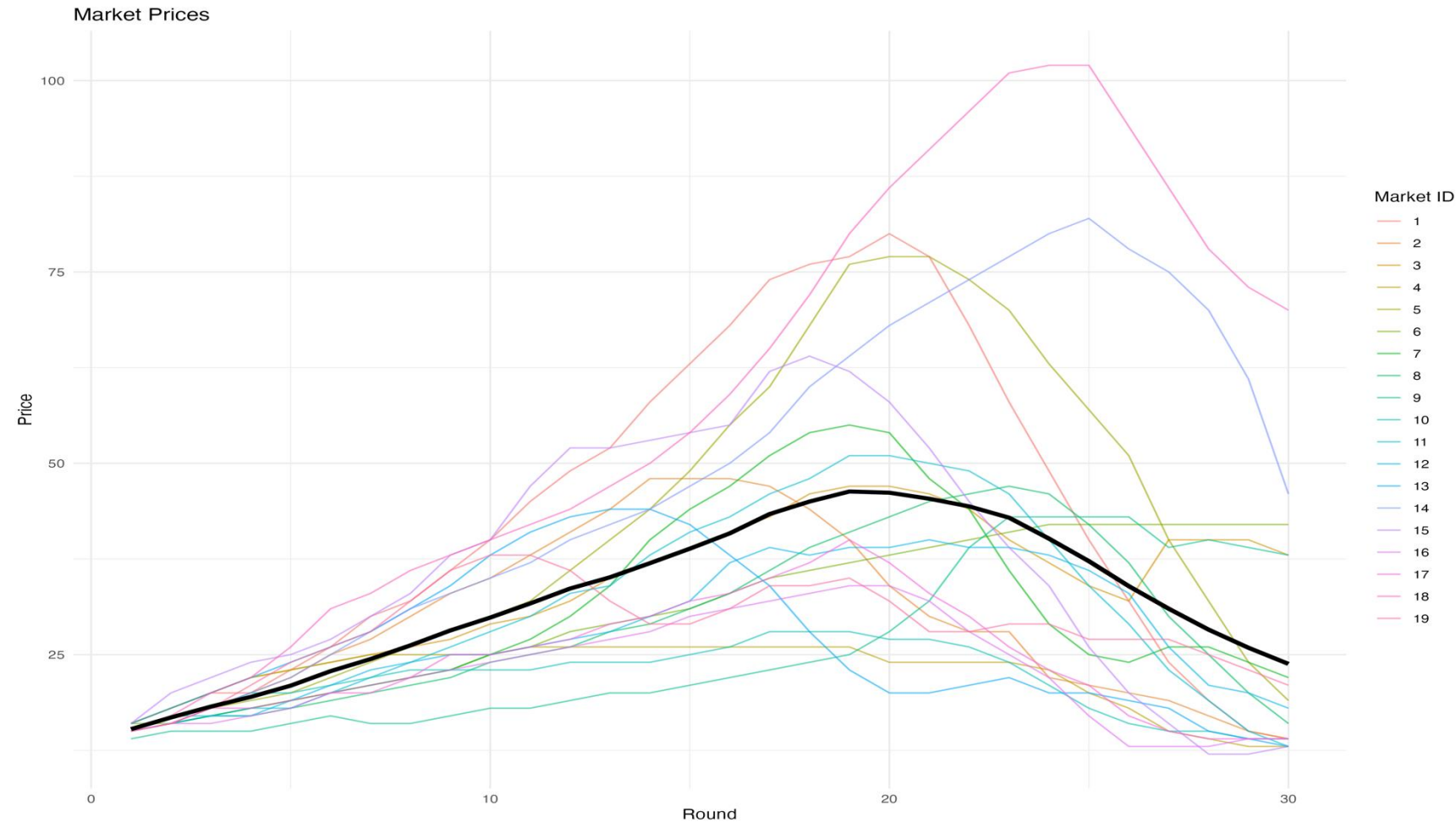


Screen 4: risk elicitation



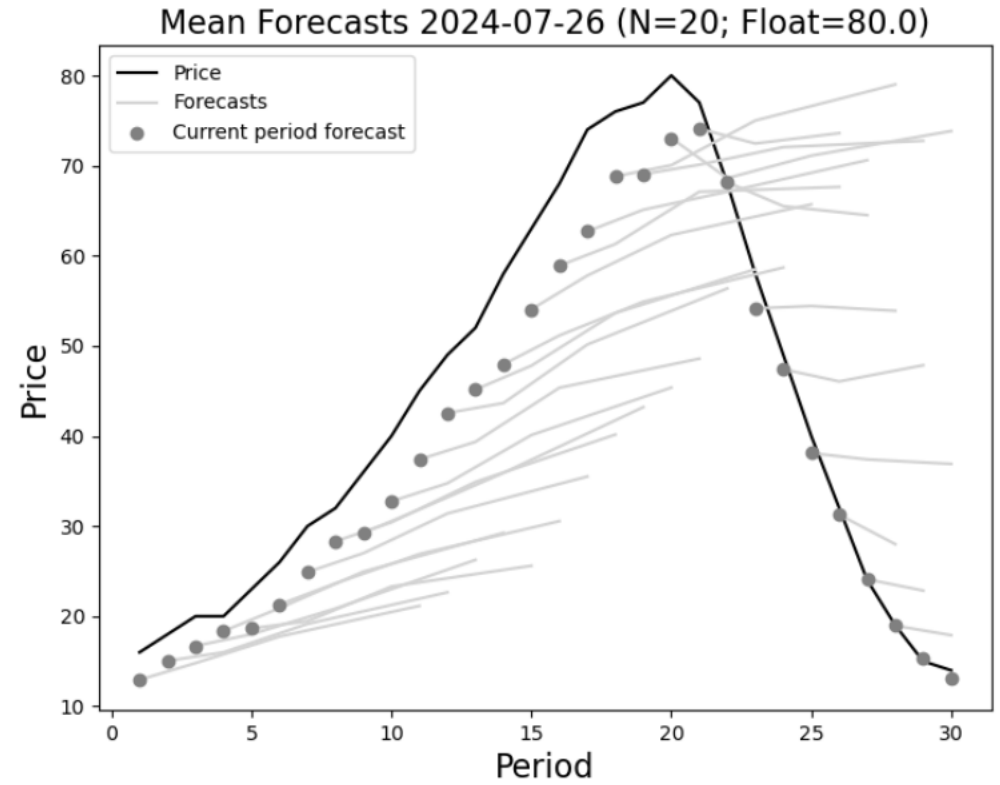
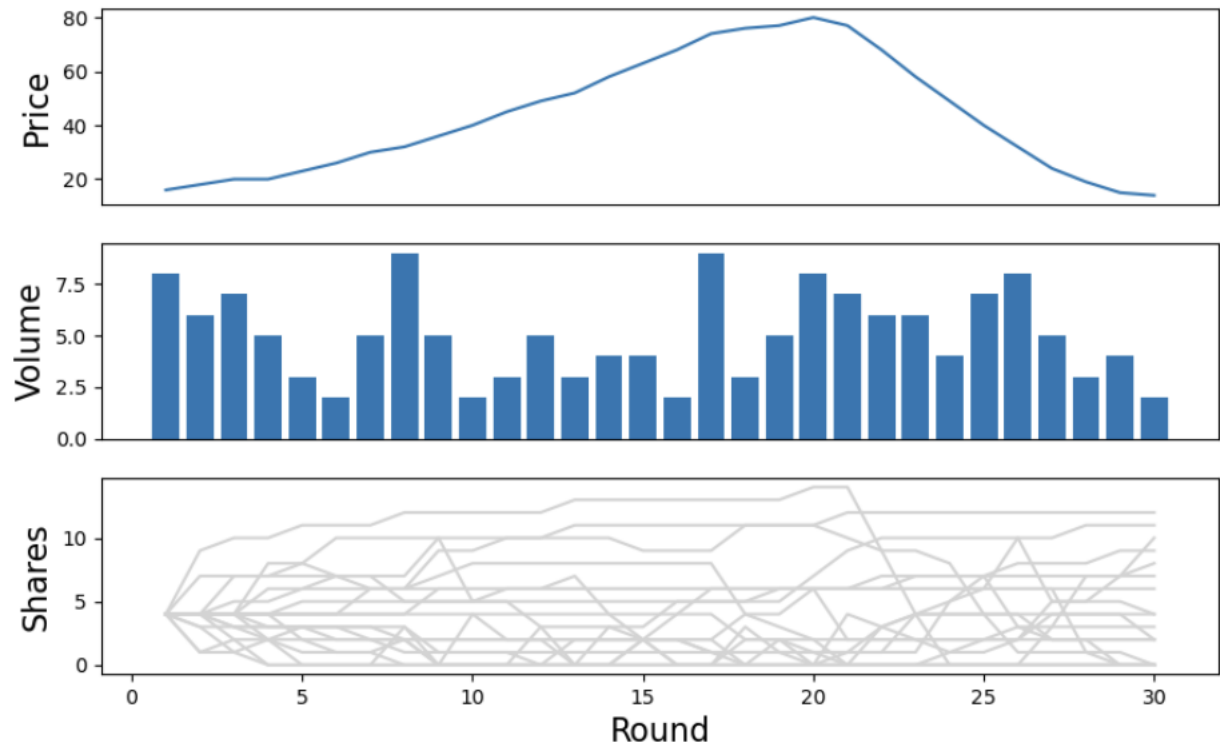
Markets overview and summary stats

- Avg. bubble deviation ~ 34
- Bubble crash ~ round 20
- Avg. volume (across rounds and markets) ~ 7 stocks

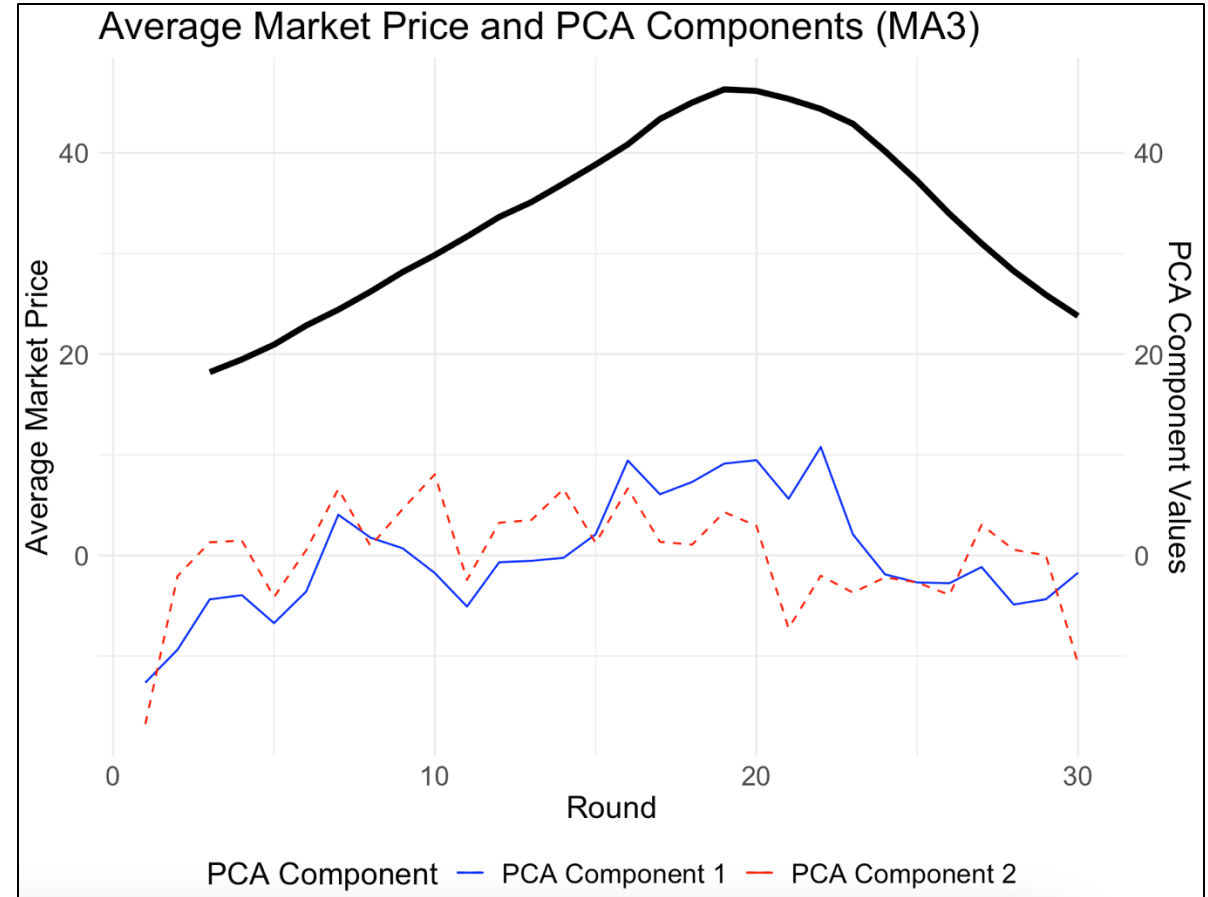
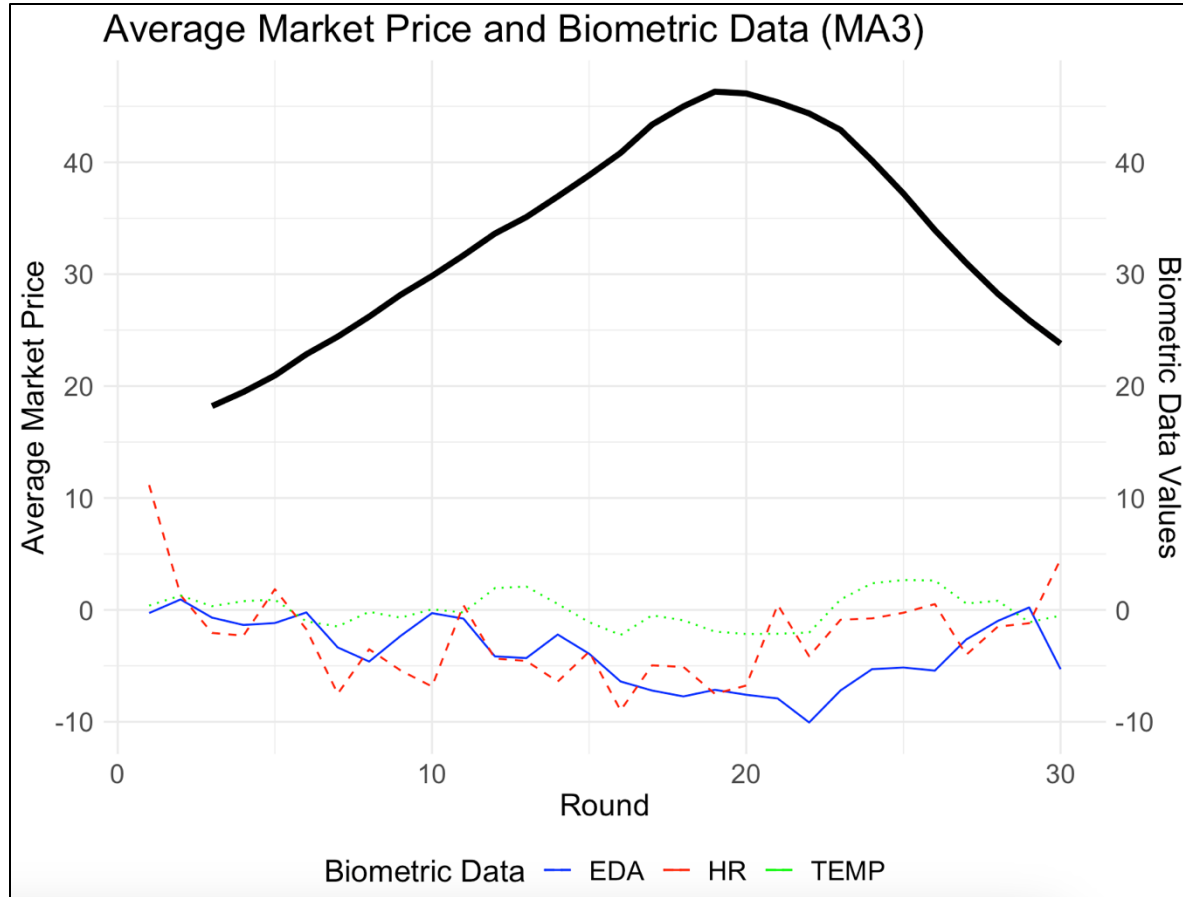


Zooming-in on a market

2024-07-26 (N = 20)



Average biodata (3-obs moving average)



Waterfall regression analysis

Model without biodata

R-Squared: 0.48426

Adj. R-Squared: **0.46091**

p-Value F-stat: < 0.00

Model with biodata

R-Squared: 0.49604

Adj. R-Squared: **0.47009**

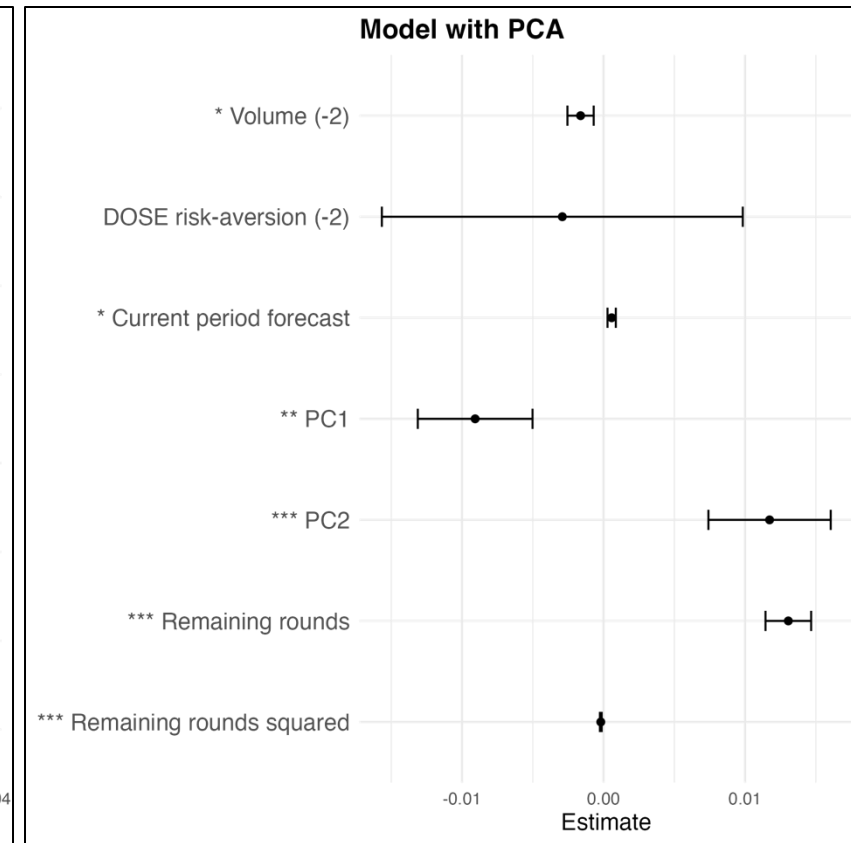
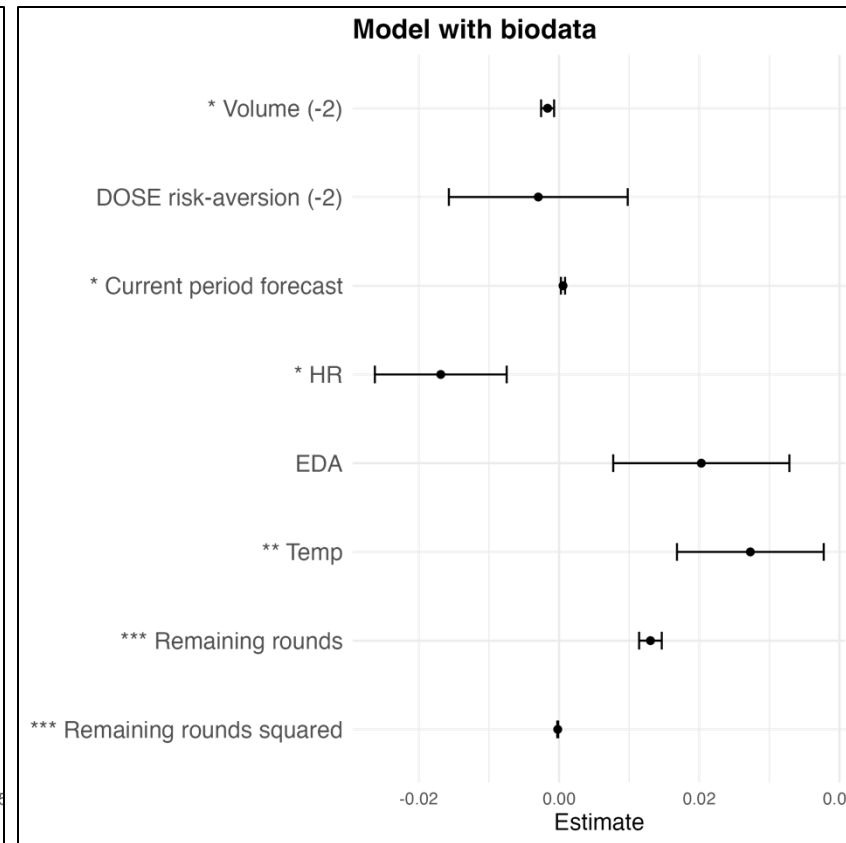
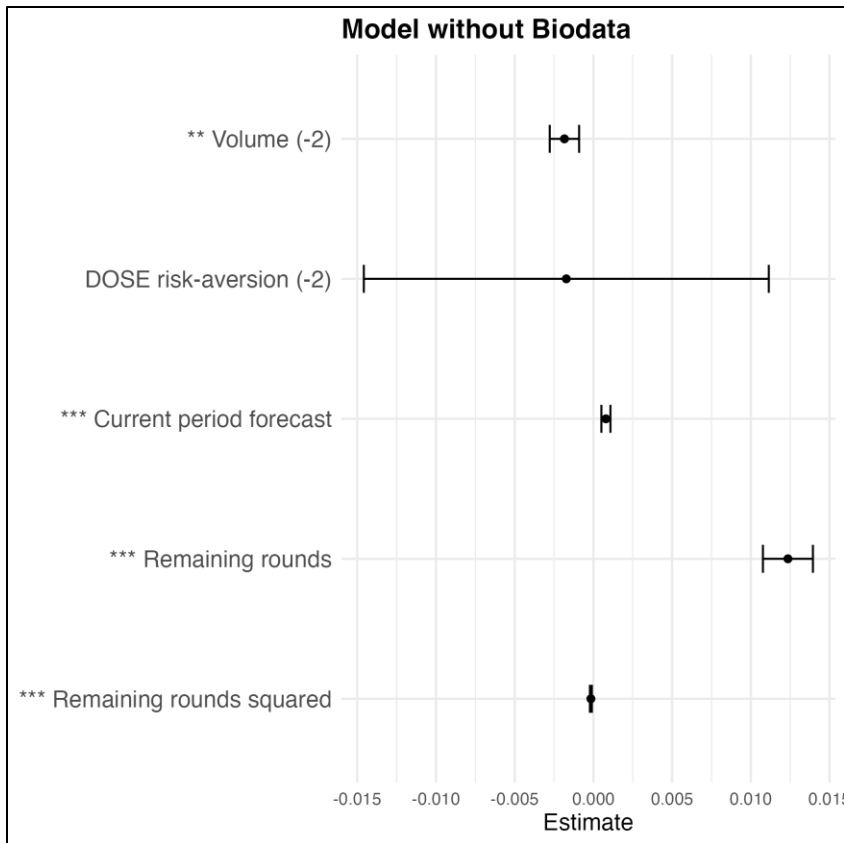
p-Value F-stat: < 0.00

Model with PCA

R-Squared: 0.49565

Adj. R-Squared: **0.47073**

p-Value F-stat: < 0.00



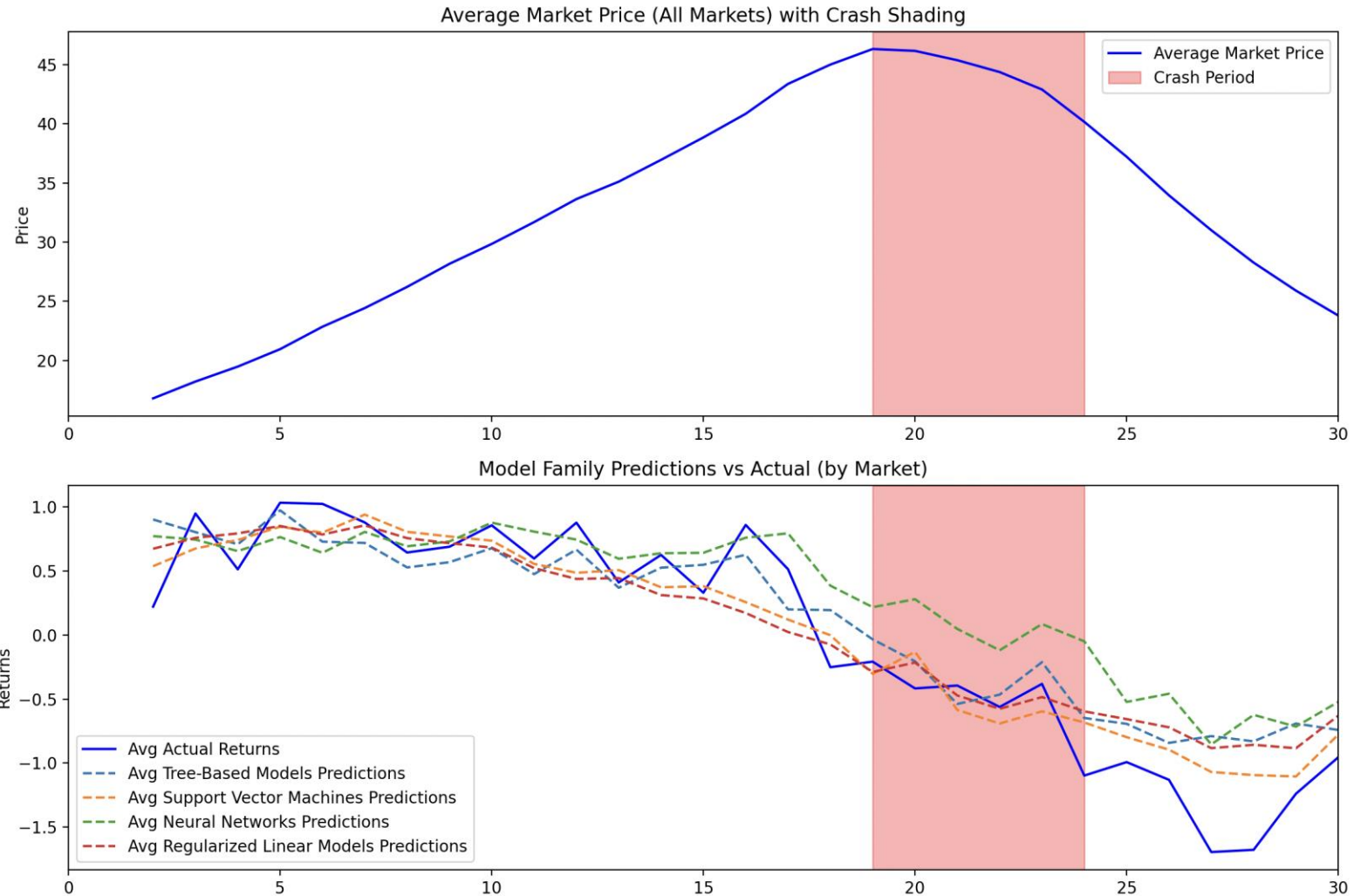
Adding biometrics/PC increases predictive power by 1 p.p.

Forecasting stock returns

Features

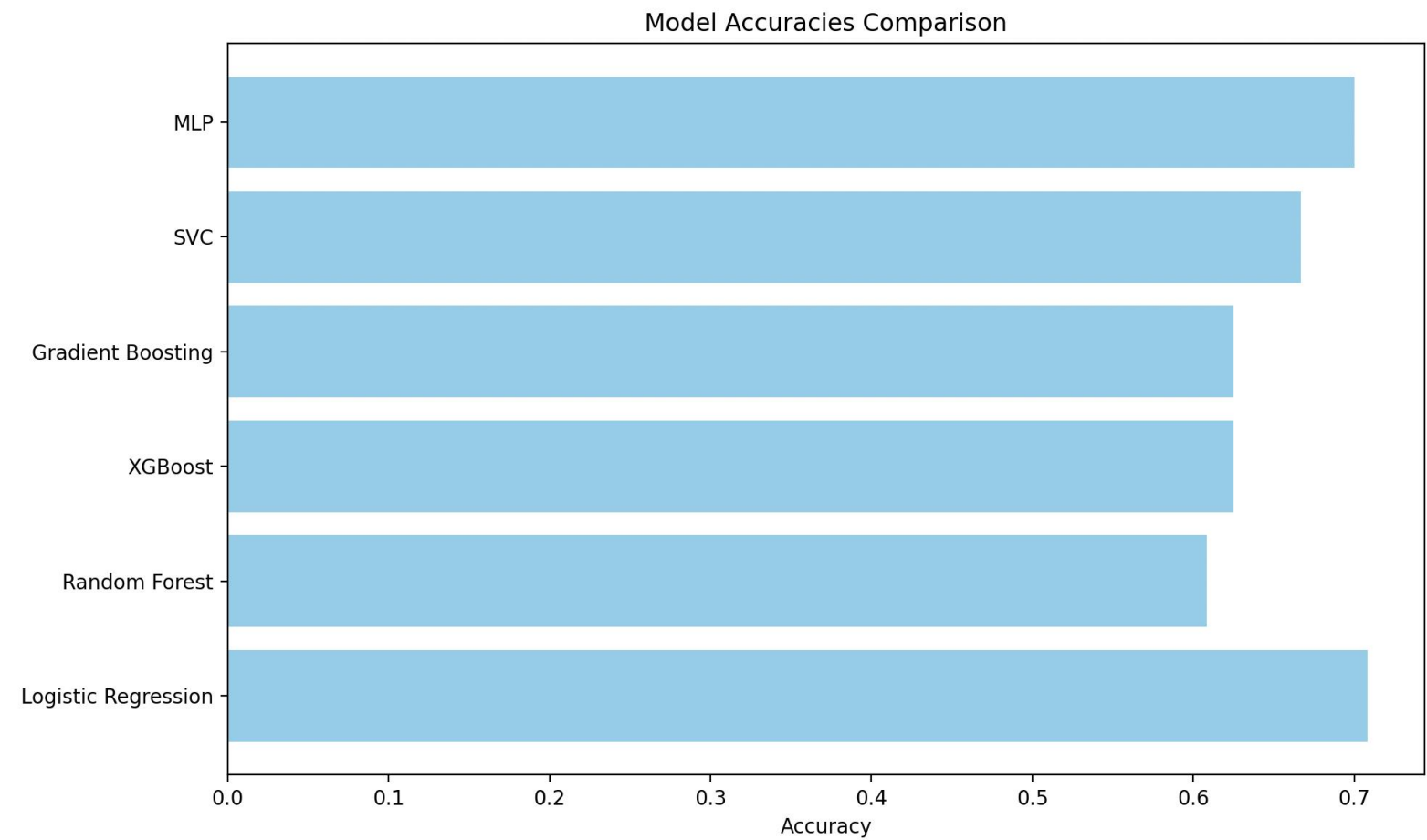
Volume, price forecasts,
risk elicitation measures,
PC components

Model	MSE	R-Squared	Expl. Var.
SVR	0.2938	69.6%	69.6%
Ridge	0.3067	68.2%	69.4%
Lasso	0.3819	60.4%	60.5%
Trees (XGBoost, Random Forest etc.)	0.4194	56.6%	58.1%
ANN, LSTM	0.5476	43.3%	48.1%



Forecasting crashes

Model family	Accuracy
Logistic	70.83%
MLP	70%
SVC	66.67%
Trees	61.94%



Conclusions

- Experimental asset markets are very useful in studying bubble dynamics – control over fundamental
- Bubbles form and crash in almost all experiments
- Individual forecasts, volume, heart rate, temperature and PC components of are predictive of trial-y-trial stock returns
- SVR/SVM and Random Forests are best for forecasting returns and crashes (further research required)

Future directions

- Further refinement of biometric data– there is no optimal consensus on pre-processing / cleaning data
- Experiments with high stakes and experienced participants (return participants)
- Fine-tune ML models to forecast crashes/stock returns based purely on biometric features
- NLP analysis on participants' feedback
- Forecasted prices analysis

Thank you!

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Selected references

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